VINOD JANGID

Computational Biologist

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SUMMARY

CSIR-NET qualified (AIR 46) Computational Biologist with specialized expertise in Structural Biology, Molecular Dynamics (MD) Simulations, and Machine Learning for drug discovery. Proven track record in rational drug design, analyzing protein-ligand interactions, and conducting free energy calculations to drive lead optimization. Proficient in developing automated Python/Bash pipelines for macromolecular modeling and high-throughput data analysis. Experienced collaborating with research teams to identify novel therapeutic targets.

EDUCATION

Master of Science in Bioinformatics

Bharathiar University | 2020 – 2022

Bachelor of Science in Biotechnology

Mohanlal Sukhadia University | 2017 – 2020

CERTIFICATIONS

- Qualified CSIR-UGC NET (Life Sciences) for Lectureship, All India Rank 46 (June 2025)
- Fellowship, Tamil Nadu State Council for Science & Technology (2022)
- Genomic Data Science Certification, Johns Hopkins University (2024)
- Computer-Aided Drug Design Certification, NPTEL, IIT Madras (2021)
- Human Molecular Genetics Certification, NPTEL, IIT Kanpur (2021)

PUBLICATION

- Jangid V., et al. (2024). Computational Analysis to Identify Novel Drug Targets for Esophageal Cancer. Medinformatics.
- Suyash S., Jangid V., et al. (2024). Machine Learning Optimization Approach to Design Multi-Epitope Marburg Vaccine Construct. Biotech Res Asia.

SKILLS

- **Structural Biology & Modeling:** AlphaFold, Rosetta, Homology Modeling, Protein-Peptide/Ligand Docking, Structure Validation, Interaction Mapping.
- **MD Simulations & Energetics:** GROMACS, AMBER, Free Energy Calculations, Complex Stability Analysis, Binding Energetics.
- **Drug Discovery & AI:** Machine Learning (ML) for Binding Prediction, QSAR Modeling, Virtual Screening, Hit-to-Lead Optimization, and Lead Identification.
- **Programming & Automation:** Python, R, Bash Scripting, Workflow Automation, High-Performance Computing (HPC), Linux/Unix.
- **Bioinformatics Tools:** PyMOL, Chimera, AutoDock, Schrödinger Suite, Sequence Analysis.

PROFESSIONAL EXPERIENCE

Computational Biologist — Jan 2025 - Present

freelance

- Conducted molecular docking, molecular dynamics (MD) simulations, and free energy calculations to identify and optimize lead compounds.
- Developed custom bioinformatics workflows for analyzing genomic and proteomic data.

Research Associate — Apr 2023 - Dec 2024

Growdea Technologies Private Limited, Gurugram

- Applied Random Forest and SVM models to predict molecular interactions from molecular descriptors, successfully identifying 3 high-potential lead compounds for a key drug discovery project.
- Designed QSAR models and automated GROMACS MD workflows, reducing simulation setup time by 40%.

Bioinformatician — Apr 2023 - March 2024

Growdea Technologies Private Limited, Gurugram

- Executed a high-throughput virtual screening campaign of over 10,000 small molecules against a novel viral target using AutoDock, prioritizing the top 0.5% of candidates for subsequent experimental validation.

Project Intern — Jan 2022 - Jun 2022

Indian Institute of Science, Bengaluru

- Conducted end-to-end RNA-Seq workflows (QC → alignment → DE analysis).
- Analyzed differentially expressed genes using GO and KEGG pathway enrichment, identifying a novel dysregulated signaling pathway implicated in omipalisib resistance in esophageal cancer cell lines.

PROJECT HIGHLIGHTS

- RNA-Seq Analysis for Oncology: Designed and executed RNA-seq workflows to identify differential gene expression in cancer datasets, including transcriptome assembly and annotation.
- Therapeutic Target Discovery: Applied QSAR and molecular docking to identify inhibitors for infectious diseases and cancer therapies.
- Drug Repurposing Studies: Conducted computational analysis to repurpose FDA-approved drugs against emerging viral threats like SARS-CoV-2 and Marburg virus.
- Network-Based Drug Discovery: Utilized network pharmacology to identify synergistic drug combinations for infectious and chronic diseases